

1. DEFINITION

DEFINITION	
Glycerol monocaprylocaprate (type I) EP / Glyceryl Mono and Dicaprylocaprate NF [NOTE – May also be labeled as USP Glyceryl Monocaprylocaprate (Type I) until May 1, 2025]	

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Glyceryl caprylate/caprate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	U72Q2I8C85
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Glycerol and Caprylic/Capric acids. No solvent used during the manufacturing process.

COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Caprylic/Capric acids (C8 and C10), the monoester fraction being predominant (Specification > 60%)	
Full Labeling (Key letters system) A₁ ≥ 75.0 ; 50.0 ≤ A₂ < 75.0 ; 25.0 ≤ B < 50.0 ; 10.0 ≤ C < 25.0 ; 5.0 ≤ D < 10.0 ; 1.0 ≤ E < 5.0 ; 0.1 ≤ F < 1.0 ; G < 0.1	
Glyceryl caprylate/caprate	A ₁ ≥ 75.0%
Other constituents (i.e. incidental ingredients)	
None	



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4. DRUG MASTER FILE

A Drug Master File (DMF) is a submission to the Food and Drug Administration (FDA) that may be used to provide confidential detailed information about facilities, processes, or articles used in the manufacturing, processing, packaging, and storing of one or more human drugs. The submission of a DMF is not required by law or FDA regulation. A DMF is submitted solely at the discretion of the holder. (Source: US FDA)

In general, the excipient DMF is not used for compendial excipients but more for novel excipients (e.g., new excipients, mixtures, coprocessed materials, and biotech excipients). FDA usually does not review DMFs for compendial excipients unless there is a specific reason to do so (e.g., new route of administration or application for existing excipients). (Source: IPEC Americas)

We have no DMF for LABRAFAC™ MC60 because it is a compendial excipient.

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopoeia Current edition	Glycerol monocaprylocaprate (type I)
USP/ NF Current edition	Glyceryl Mono and Dicaprylocaprate

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓(5) GRAS 184.1505		

EUROPEAN REGULATION

(1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)

(2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units)

(3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether)

(4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin

USA REGULATION

- Veterinary drugs

FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume.

In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety

(5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)

(6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)

- Pesticides

(8) EPA's pesticides program (<http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::>)



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7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	glyceryl caprylate/caprate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Over the Counter, Prescription Medicines Not available as an Equivalent Ingredient in any application	Please note: Only the name and definition of a substance have been reviewed to allow it to be included in the ingredient repository. The approval for use of the ingredient in therapeutic goods is a decision made by the relevant TGA regulatory area. This approval process may require submission of further information, for example safety data for the ingredient or for the finished goods, to meet legislative and regulatory requirements.
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved non-medicinal ingredient	Glyceryl Caprylate/Caprate	Topical use
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	Reference	Maximum "as used" concentration	Conclusions
Monoglyceryl monoesters	CIR report, 2016	No information available	The CIR Expert Panel concluded that the following ingredients are safe in cosmetics in the present practices of use and concentration described in this safety assessment.



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9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
	CAS #	EC #	Chemical name
Glyceryl caprylate/caprate	85536-07-8 (or 91744-32-0)	287-488-0 (or 294-594-0)	Glycerides, C8-10 mono- and di- (or Glycerides, C8-10 mono-, di- and tri-)

NATIONAL INVENTORIES					
	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIC)	China (IECSC)	Other inventories
Glyceryl caprylate/caprate	Not listed (or Not listed)	Not listed (or Not listed)	Listed (or not listed)	Listed Listed	C8-10 mixed mono-and diglyceride: IECSC / Glycerides, C8-10 mono- and di-: AIC, EINECS, NZIoC, REACH, TCSI, VNECI / Glycerides, (C8-10) mono- and di-: AREC, ECL (or Glycerides, C8-10 mono-, di-and tri-: IECSC, EINECS, PICCS, REACH, TCSI, VNECI / Glycerides, (C8-10) mono-, di- and tri-: AREC, ECL)
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting. Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according to Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	Status per substance
Glyceryl caprylate/caprate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.



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10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Glyceryl Mono and Dicaprylocaprate	Oral	Capsule	765 mg	
Glyceryl Mono and Dicaprylocaprate	Oral	Capsule, extended release		24 mg
Glyceryl Mono and Dicaprylocaprate	Oral	Capsule, Gelatin coated	62.2 mg	
Glyceryl Mono and Dicaprylocaprate	Oral	Capsule, Liquid filled	139 mg	
Glyceryl Mono and Dicaprylocaprate	Oral	Solution	30 mg / 1 ml	
Glyceryl Mono and Dicaprylocaprate	Oral	Tablet	1.3 mg	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Glyceryl monocaprylocaprate (UNII: G7515SW10N)	Oral	Capsule		502 mg
Glyceryl monocaprylate (UNII: TM2TZD4G4A)	Oral	Capsule	400 mg	
	Oral	Solution	349.1 mg/mL	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.3. FDA IIG references for larger family of chemicals

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Medium-chain triglycerides (UNII: C9H2L21V7U)	INTRAVENOUS	INJECTION, EMULSION	1.1%w/v	
Medium-chain triglycerides (UNII: C9H2L21V7U)	ORAL	CAPSULE		4771mg
	ORAL	CAPSULE, EXTENDED RELEASE	3.08mg	
	ORAL	CAPSULE, LIQUID FILLED		866mg
	ORAL	POWDER, FOR SUSPENSION	3390mg/5ml	
	ORAL	SOLUTION		946mg
	ORAL	SUSPENSION		10170mg
	ORAL	TABLET		2mg
	ORAL	TABLET, EXTENDED RELEASE	0.37mg	
	ORAL	TABLET, ORALLY DISINTEGRATING		1mg
Medium-chain triglycerides (UNII: C9H2L21V7U)	SUBCUTANEOUS	INJECTION, SUSPENSION, EXTENDED RELEASE	990.3mg	
Medium-chain triglycerides (UNII: C9H2L21V7U)	SUBLINGUAL	POWDER		2mg
	SUBLINGUAL	SPRAY METERED		110mg
Medium-chain triglycerides (UNII: C9H2L21V7U)	TOPICAL	AEROSOL, FOAM	10.87%w/w	



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
	TOPICAL	CREAM		570mg
	TOPICAL	CREAM, AUGMENTED	8%w/w	
	TOPICAL	GEL	10%w/w	
	TOPICAL	LOTION	7%w/w	
	TOPICAL	OINTMENT	20%w/w	
	TOPICAL	SPRAY		540mg
Medium-chain triglycerides (UNII: C9H2L21V7U)	TRANSDERMAL	FILM, EXTENDED RELEASE	NA	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.2. Food/Nutraceutical/Dietary supplement applications

Mono- and Diglycerides have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration identifier
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides
U.S.A.	GRAS US Food additives according to 21 CFR	21CFR §184.1505 Mono- and Diglycerides
JAPAN	JSFA (Japanese Standards of Food Additives)	Glycerol esters of fatty acids

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

The information contained in this document is based on our knowledge of the related product(s) at the date indicated herein, and is provided in good faith. Notwithstanding the foregoing, no guarantee is made in respect of the product's(s) properties or efficacy, or the continued exactitude and completeness of the information herein. No contractual tie is formed by the provision of this information. This document does not relieve the user from conducting all tests and verifications to ensure the product's(s) suitability for each intended use. Nothing herein shall be construed as a recommendation or licence to use any information that does or may conflict with any intellectual property. We make no representation or warranty, express or implied, that use of this information will not infringe any intellectual property of Gattefossé or any third party. The user assumes all risks and liability incurred if a product is used other than for the purpose designated by Gattefossé. This statement does not relieve the user from verifying its conformity with all current applicable laws and regulations, and the user shall remain solely liable for ensuring its continued conformity therewith.

